User Input — Raw Dataset + Natural-Language User Goal

Phase 1 — Intent Parsing & Task Classification
(target column, task type, metric resolution)

Phase 2 — Preprocessing Planning
(null strategy, encoding, scaling plan)

Phase 3 — Data Cleaning & Feature Engineering

re-wind to failed step if preprocessing output is invalid

Basic Cleaning | Train-Test Split | EDA | Feature Selection | Deep Cleaning | Feature Engineering | Verification (rule-based check)

Phase 4 — Execution Planning
(model shortlist, search strategy, CV design, threshold)

Phase 5 — Model Training & Evaluation
Progressive elimination → Optuna TPE + LLM HPO priors →
diversity-aware ensembling → threshold optimization

Phase 6 — Threshold Check

Pass

Phase 7 — Report Generation & Final Output
(stakeholder report, model recommendation, pipeline code)

Read and write decisions

Episodic Decision Log
Every routed decision + rationale, timestamped

3-Tier Decision Router

Tier 1: Rule-based

Tier 2: Statistical

Tier 3: LLM

re-enters Phase 3 or 4

Fail

Phase 6 — Failure Diagnosis
(data / feature / model /
fundamental-insufficiency)

Phase 6 — Targeted Revision
(re-enters affected phase only)